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Element A

Team Towers is a team of Brentwood High School Engineering focused on creating a product that is both functional and effective for a cart extension. Involved in this project are a series of students and mentors, including Nicolas Blucher, Connor Harris, and Bennett Lohmeyer along with Shanti Boorgu and Ellison Abramowitz representing the people under or around 5'. The team has taken several samples from clients of several different heights, but the primary stakeholder is Mr.Kiser who this design is being built around. Within the school day and sometimes over the weekend when administering tests, Mr.Kiser pushes a cart around the school while bending over farther than preferable. The challenge is creating a device that is suitable for people ranging from all heights 5'8" - 6'5".

Below we have images depicting the client's problem:

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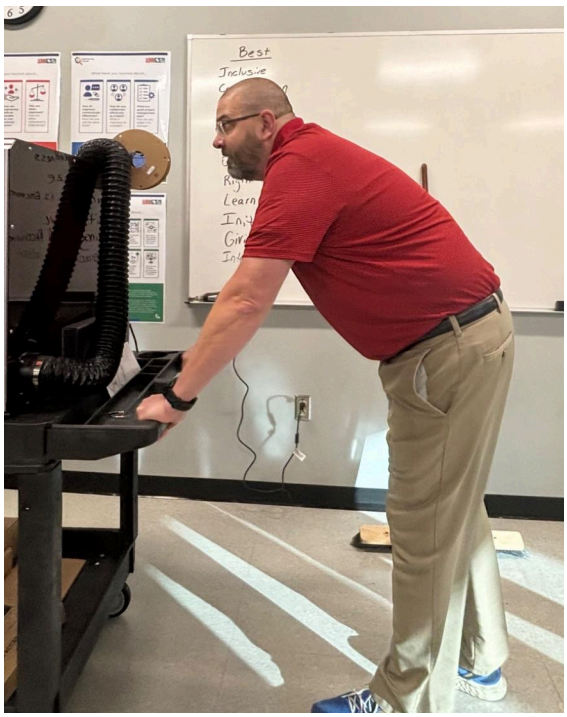

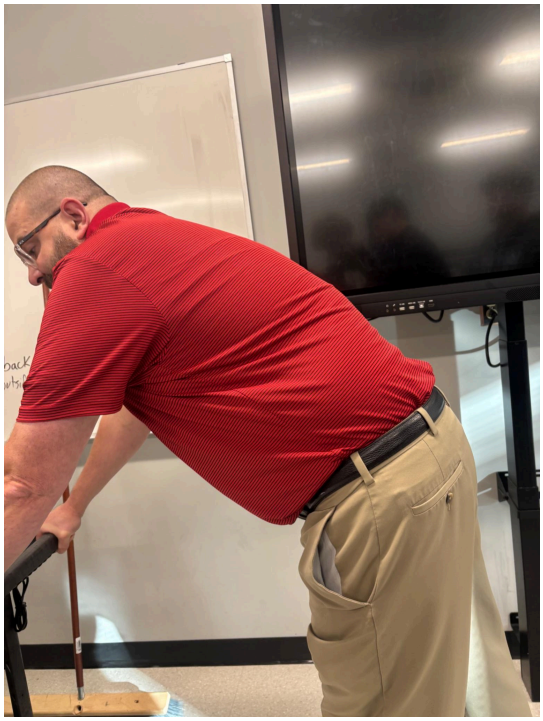

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Although this issue is not a universal problem the implementation of this device has no limitations. As of right now, the only client we currently have is Mr.Kiser; however, with the right design and functionality, this product could be utilized in schools nationally where these carts are shipped. The biggest problem is a result of the height of the client in comparison to the height of the cart's handle. Because of this, Mr.Kiser is forced to adjust to uncomfortable measures to push the cart. On top of that, these carts sometimes carry up to 500 pounds of weight which forces the person pushing the cart to apply a large amount of force to get the cart to move. Using this cart multiple times a week, carrying textbooks, standardized tests, and packages can become quite painful having to overextend your back. This problem is fairly common among cart manufacturers because this is a minimal problem.

The primary concern is the back strain from the constant manual labor of pushing and pulling the cart. On the other hand, the designer's and manufacturers' main concern is the functionality and ergonomics of the design to fit the consumer's wants. The need to address this problem is necessary because it would mitigate the opportunity for chronic back pain while also allowing for prime productivity. Applied Ergonomics, Human Factors: The Journal of the Human Factors and Ergonomics Society, and Journal of Biomechanics were the available peer-reviewed journal articles that we were able to find. Based on these

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we were able to further understand the growing issue and those that were affected by the issue at hand.

Below are images depicting the issues and the desired height that the client wants along with the positioning of his back:

The current issue lies in the lumbar because the force exerted requires the user to lean upwards of 45° to have enough force to move the cart, putting extra weight on his lower back to compensate for the weight of the cart and generate enough force to move the cart.

The product must meet several requirements set by Mr.Kiser to be considered a success. The design must have some sort of adjustability so that it can vary from height to height and consistently have the same functionality. The next standard set by the client was that

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the handle must have some ergonomic aspect to the design. For comfort and aesthetics, the design must be ergonomically fit for the hands of the user in a universal sense. On top of all this, the product must also be under a budget of $30. This is a very impactful budget because it prevents too much overcomplication, yet promotes simplicity and functionality. The problem is that in order to create a functional design with all these design requirements strong would go a long way to helping achieve those requirements. The last requirement lies in the durability and long-lasting aspect of the design. The design must last up to 5 years probably even longer than that and must still be functional for day-to-day use. Other designs have yet to succeed in something that meets all these criteria. This is because this is a very minuscule problem in most people's eyes; however, for the users this could lead to a lifetime of chronic back pain and prevention of certain activities with growing age.

Some stakeholders and stakeholder groups are directly impacted by this problem. Certain teachers who use these carts, specifically Mr.Kiser, are directly affected by this problem. Several others are also impacted by this problem including ergonomic designers, manufacturers and distributors, users in manual material handling, and certain healthcare providers. All these different stakeholders are impacted by this problem in a different way but ultimately lead to the same source/issue. Indeed there are defects and flaws in the design, but the problem must be addressed and solved. Secondary stakeholders aren't

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extremely relevant in terms of the scope of this project because the groups most affected by this problem are the priority.

In previous years several other sources have attempted to or designed something in a similar fashion. Applied Ergonomics, Journal of Biomechanics, and Human Factors are just a few of the credible sources that address this problem and specifically explain their process in the peer-reviewed journals. Applied Ergonomics is a website specifically focused on the ergonomic feel of designs. They process and judge the ergonomic ability of the object and explain themselves in the peer-reviewed journals that follow: Applied Ergonomics Source 1 and Applied Ergonomics Source 2. The next credible source is the Journal of Biomechanics whose focus is on the biomechanical side, which emphasizes designing and developing devices related to the human body. This is the source with the most focus on more human impact and factors. In short biomechanical engineering combines principles of biological knowledge and mechanical engineering to understand how to improve peoples' quality of life. Journal of Biomechanics Source 1 and Journal of Biomechanics Source 2 both touch on the biomechanical side of the design process and explain the human factor side of the design. It touches on the muscular impact along with validation as to a design focused on pushing and pulling. The last and final source found was from the Human Factors: The Journal of the Human Factors and Ergonomics Society

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discusses ideas on how to reduce spinal load on the lumbar spine and the human and ergonomics behind the design.

Below are the instructions for putting the cart together as well as the parts necessary for construction:

7. Use as intended only.
8. Inspect before every use; do not use if parts are loose or damaged.

ASSEMBLY

Use the mallet to tap down the parts to ensure they are seated properly before fastening them with screws.

1 .Turn upper platform upright (#2) and install handle (#1)
2.Turn upper platform assembly upside down, and install legs (twoleft and two right, #4and#5）
3.install lower platform(#3) and castors(#6 #7） Tighten all screws and hex nuts. (#8, #9, #10)

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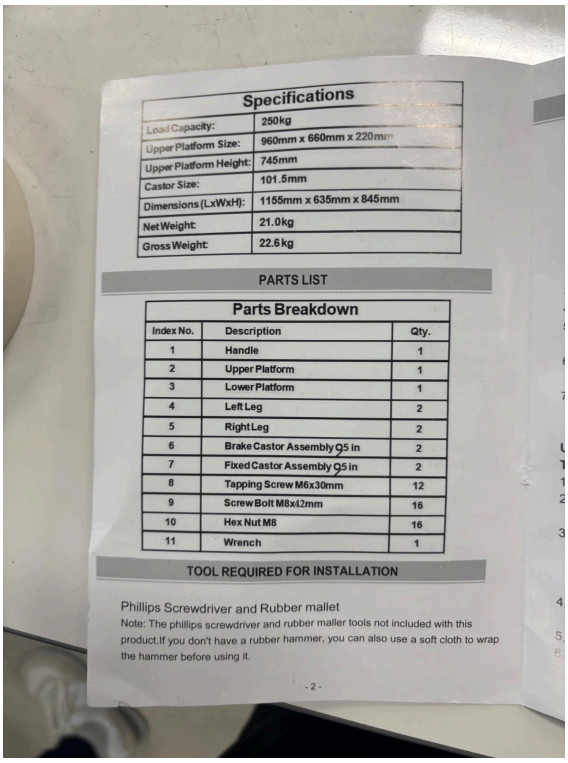

Specifications

Load Capacity:	250kg
Upper Platform Size:	960mm x 660mm x 220mm
Upper Platform Height:	745mm
Castor Size:	101.5mm
Dimensions (LxWxH):	1155mm x 635mm x 845mm
Net Weight:	21.0kg
Gross Weight:	22.6kg

PARTS LIST

Parts Breakdown

Index No.	Description	Qty.
1	Handle	1
2	Upper Platform	1
3	Lower Platform	1
4	Left Leg	2
5	Right Leg	2
6	Brake Castor Assembly Q5 in	2
7	Fixed Castor Assembly Q5 in	2
8	Tapping Screw M6x30mm	12
9	Screw Bolt M8xl2mm	16
10	Hex Nut M8	16
11	Wrench	1

TOOL REQUIRED FOR INSTALLATION

Phillips Screwdriver and Rubber mallet

Note: The phillips screwdriver and rubber maller tools not included with this product.If you don't have a rubber hammer, you can also use a soft cloth to wrap the hammer before using it.

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Above are the specifications that we must work with to create a design that will suffice for this cart as well as others that the client may use. By using the specs provided we were able to calculate the heights and specifications that we would need for our design to be functional and work the way we want.